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NATURAL TTS SYNTHESIS BY CONDITIONING WAVENET ON MEL SPECTROGRAM PREDICTIONS

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ABSTRACT

This paper describes Tacotron 2, a neural network architecture for speech synthesis directly from text. The system is composed of a recurrent sequence-to-sequence feature prediction network that maps character embeddings to mel-scale spectrograms, followed by a modified WaveNet model acting as a vocoder to synthesize time-domain waveforms from those spectrograms. Our model achieves a mean opinion score (MOS) of 4.53 comparable to a MOS of 4.58 for professionally recorded speech. To validate our design choices, we present ablation studies of key components of our system and evaluate the impact of using mel spectrograms as the conditioning input to WaveNet instead of linguistic, duration, and F_0 features. We further show that using this compact acoustic intermediate representation allows for a significant reduction in the size of the WaveNet architecture.

Index Terms— Tacotron 2, WaveNet, text-to-speech

1. INTRODUCTION

Generating natural speech from text (text-to-speech synthesis, TTS) remains a challenging task despite decades of investigation [1]. Over time, different techniques have dominated the field. Concatenative synthesis with unit selection, the process of stitching small units of pre-recorded waveforms together [2, 3] was the state-of-the-art for many years. Statistical parametric speech synthesis [4, 5, 6, 7], which directly generates smooth trajectories of speech features to be synthesized by a vocoder, followed, solving many of the issues that concatenative synthesis had with boundary artifacts. However, the audio produced by these systems often sounds muffled and unnatural compared to human speech.

WaveNet [8], a generative model of time domain waveforms, produces audio quality that begins to rival that of real human speech and is already used in some complete TTS systems [9, 10, 11]. The inputs to WaveNet (linguistic features, predicted log fundamental frequency (F_0), and phoneme durations), however, require significant domain expertise to produce, involving elaborate text-analysis systems as well as a robust lexicon (pronunciation guide).

Tacotron [12], a sequence-to-sequence architecture [13] for producing magnitude spectrograms from a sequence of characters, simplifies the traditional speech synthesis pipeline by replacing the production of these linguistic and acoustic features with a single neural network trained from data alone. To vocode the resulting magnitude spectrograms, Tacotron uses the Griffin-Lim algorithm [14] for phase estimation, followed by an inverse short-time Fourier transform. As the authors note, this was simply a placeholder for future neural vocoder approaches, as Griffin-Lim produces characteristic artifacts and lower audio quality than approaches like WaveNet.

In this paper, we describe a unified, entirely neural approach to speech synthesis that combines the best of the previous approaches: a sequence-to-sequence Tacotron-style model [12] that generates mel spectrograms, followed by a modified WaveNet vocoder [10, 15]. Trained directly on normalized character sequences and corresponding speech waveforms, our model learns to synthesize natural sounding speech that is difficult to distinguish from real human speech.

Deep Voice 3 [11] describes a similar approach. However, unlike our system, its naturalness has not been shown to rival that of human speech. Char2Wav [16] describes yet another similar approach to end-to-end TTS using a neural vocoder. However, they use different intermediate representations (traditional vocoder features) and their model architecture differs significantly.

2. MODEL ARCHITECTURE

Our proposed system consists of two components, shown in Figure 1: (1) a recurrent sequence-to-sequence feature prediction network with attention which predicts a sequence of mel spectrogram frames from an input character sequence, and (2) a modified version of WaveNet which generates time-domain waveform samples conditioned on the predicted mel spectrogram frames.

2.1. Intermediate Feature Representation

In this work we choose a low-level acoustic representation: mel-frequency spectrograms, to bridge the two components. Using a representation that is easily computed from time-domain waveforms allows us to train the two components separately. This representation is also smoother than waveform samples and is easier to train using a squared error loss because it is invariant to phase within each frame.

A mel-frequency spectrogram is related to the linear-frequency spectrogram, i.e., the short-time Fourier transform (STFT) magnitude. It is obtained by applying a nonlinear transform to the frequency axis of the STFT, inspired by measured responses from the human auditory system, and summarizes the frequency content with fewer dimensions. Using such an auditory frequency scale has the effect of emphasizing details in lower frequencies, which are critical to speech intelligibility, while de-emphasizing high frequency details, which are dominated by fricatives and other noise bursts and generally do not need to be modeled with high fidelity. Because of these properties, features derived from the mel scale have been used as an underlying representation for speech recognition for many decades [17].

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While linear spectrograms discard phase information (and are therefore lossy), algorithms such as Griffin-Lim [14] are capable of estimating this discarded information, which enables time-domain conversion via the inverse short-time Fourier transform. Mel spectrograms discard even more information, presenting a challenging inverse problem. However, in comparison to the linguistic and acoustic features used in WaveNet, the mel spectrogram is a simpler, lower-level acoustic representation of audio signals. It should therefore be straightforward for a similar WaveNet model conditioned on mel spectrograms to generate audio, essentially as a neural vocoder. Indeed, we will show that it is possible to generate high quality audio from mel spectrograms using a modified WaveNet architecture.

2.2. Spectrogram Prediction Network

As in Tacotron, mel spectrograms are computed through a short-time Fourier transform (STFT) using a 50 ms frame size, 12.5 ms frame hop, and a Hann window function. We experimented with a 5 ms frame hop to match the frequency of the conditioning inputs in the original WaveNet, but the corresponding increase in temporal resolution resulted in significantly more pronunciation issues.

We transform the STFT magnitude to the mel scale using an 80 channel mel filterbank spanning 125 Hz to 7.6 kHz, followed by log dynamic range compression. Prior to log compression, the filterbank output magnitudes are clipped to a minimum value of 0.01 in order to limit dynamic range in the logarithmic domain.

The network is composed of an encoder and a decoder with attention. The encoder converts a character sequence into a hidden feature representation which the decoder consumes to predict a spectrogram. Input characters are represented using a learned 512-dimensional character embedding, which are passed through a stack of 3 convolutional layers each containing 512 filters with shape 5×1, i.e., where each filter spans 5 characters, followed by batch normalization [18] and ReLU activations. As in Tacotron, these convolutional layers model longer-term context (e.g., N-grams) in the input character sequence. The output of the final convolutional layer is passed into a single bi-directional [19] LSTM [20] layer containing 512 units (256 in each direction) to generate the encoded features.

The encoder output is consumed by an attention network which summarizes the full encoded sequence as a fixed-length context vector for each decoder output step. We use the location-sensitive attention from [21], which extends the additive attention mechanism [22] to use cumulative attention weights from previous decoder time steps as an additional feature. This encourages the model to move forward consistently through the input, mitigating potential failure modes where some subsequences are repeated or ignored by the decoder. Attention probabilities are computed after projecting inputs and location features to 128-dimensional hidden representations. Location features are computed using 32 1-D convolution filters of length 31.

The decoder is an autoregressive recurrent neural network which predicts a mel spectrogram from the encoded input sequence one frame at a time. The prediction from the previous time step is first passed through a small *pre-net* containing 2 fully connected layers of 256 hidden ReLU units. We found that the pre-net acting as an information bottleneck was essential for learning attention. The pre-net output and attention context vector are concatenated and passed through a stack of 2 uni-directional LSTM layers with 1024 units. The concatenation of the LSTM output and the attention context vector is projected through a linear transform to predict the target spectrogram frame. Finally, the predicted mel spectrogram is passed through a 5-layer convolutional *post-net* which predicts a residual to add to the prediction to improve the overall reconstruction. Each

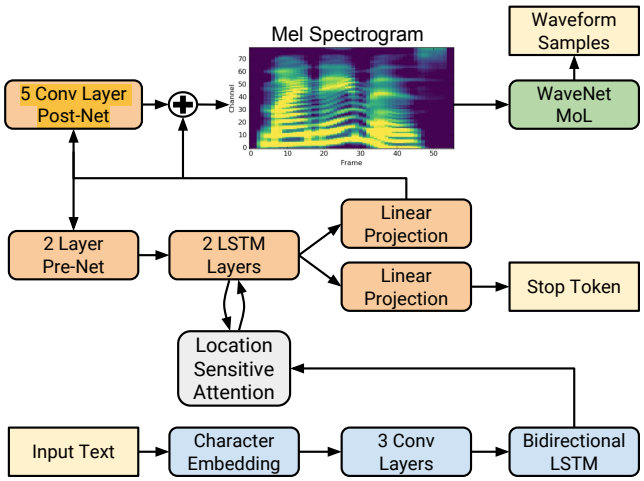

Fig. 1. Block diagram of the Tacotron 2 system architecture.

post-net layer is comprised of 512 filters with shape 5×1 with batch normalization, followed by $\tanh$ activations on all but the final layer.

We minimize the summed mean squared error (MSE) from before and after the post-net to aid convergence. We also experimented with a log-likelihood loss by modeling the output distribution with a Mixture Density Network [23, 24] to avoid assuming a constant variance over time, but found that these were more difficult to train and they did not lead to better sounding samples.

In parallel to spectrogram frame prediction, the concatenation of decoder LSTM output and the attention context is projected down to a scalar and passed through a sigmoid activation to predict the probability that the output sequence has completed. This "stop token" prediction is used during inference to allow the model to dynamically determine when to terminate generation instead of always generating for a fixed duration. Specifically, generation completes at the first frame for which this probability exceeds a threshold of 0.5.

The convolutional layers in the network are regularized using dropout [25] with probability 0.5, and LSTM layers are regularized using zoneout [26] with probability 0.1. In order to introduce output variation at inference time, dropout with probability 0.5 is applied only to layers in the pre-net of the autoregressive decoder.

In contrast to the original Tacotron, our model uses simpler building blocks, using vanilla LSTM and convolutional layers in the encoder and decoder instead of "CBHG" stacks and GRU recurrent layers. We do not use a "reduction factor", i.e., each decoder step corresponds to a single spectrogram frame.

2.3. WaveNet Vocoder

We use a modified version of the WaveNet architecture from [8] to invert the mel spectrogram feature representation into time-domain waveform samples. As in the original architecture, there are 30 dilated convolution layers, grouped into 3 dilation cycles, i.e., the dilation rate of layer k ($k = 0 \ldots 29$) is $2^{k \pmod{10}}$. To work with the 12.5 ms frame hop of the spectrogram frames, only 2 upsampling layers are used in the conditioning stack instead of 3 layers.

Instead of predicting discretized buckets with a softmax layer, we follow PixelCNN++ [27] and Parallel WaveNet [28] and use a 10-component mixture of logistic distributions (MoL) to generate 16-bit samples at 24 kHz. To compute the logistic mixture distribution, the WaveNet stack output is passed through a ReLU activation followed

by a linear projection to predict parameters (mean, log scale, mixture weight) for each mixture component. The loss is computed as the negative log-likelihood of the ground truth sample.

3. EXPERIMENTS & RESULTS

3.1. Training Setup

Our training process involves first training the feature prediction network on its own, followed by training a modified WaveNet independently on the outputs generated by the first network.

To train the feature prediction network, we apply the standard maximum-likelihood training procedure (feeding in the correct output instead of the predicted output on the decoder side, also referred to as *teacher-forcing*) with a batch size of 64 on a single GPU. We use the Adam optimizer [29] with $\beta_1 = 0.9, \beta_2 = 0.999, \epsilon = 10^{-6}$ and a learning rate of 10^{-3} exponentially decaying to 10^{-5} starting after 50,000 iterations. We also apply L_2 regularization with weight 10^{-6}.

We then train our modified WaveNet on the *ground truth-aligned* predictions of the feature prediction network. That is, the prediction network is run in teacher-forcing mode, where each predicted frame is conditioned on the encoded input sequence and the corresponding previous frame in the ground truth spectrogram. This ensures that each predicted frame exactly aligns with the target waveform samples.

We train with a batch size of 128 distributed across 32 GPUs with synchronous updates, using the Adam optimizer with $\beta_1 = 0.9, \beta_2 = 0.999, \epsilon = 10^{-8}$ and a fixed learning rate of 10^{-4}. It helps quality to average model weights over recent updates. Therefore we maintain an exponentially-weighted moving average of the network parameters over update steps with a decay of 0.9999 – this version is used for inference (see also [29]). To speed up convergence, we scale the waveform targets by a factor of 127.5 which brings the initial outputs of the mixture of logistics layer closer to the eventual distributions.

We train all models on an internal US English dataset[12], which contains 24.6 hours of speech from a single professional female speaker. All text in our datasets is spelled out. e.g., "16" is written as "sixteen", i.e., our models are all trained on normalized text.

3.2. Evaluation

When generating speech in inference mode, the ground truth targets are not known. Therefore, the predicted outputs from the previous step are fed in during decoding, in contrast to the teacher-forcing configuration used for training.

We randomly selected 100 fixed examples from the test set of our internal dataset as the evaluation set. Audio generated on this set are sent to a human rating service similar to Amazon's Mechanical Turk where each sample is rated by at least 8 raters on a scale from 1 to 5 with 0.5 point increments, from which a subjective mean opinion score (MOS) is calculated. Each evaluation is conducted independently from each other, so the outputs of two different models are not directly compared when raters assign a score to them.

Note that while instances in the evaluation set never appear in the training set, there are some recurring patterns and common words between the two sets. While this could potentially result in an inflated MOS compared to an evaluation set consisting of sentences generated from random words, using this set allows us to compare to the ground truth. Since all the systems we compare are trained on the same data, relative comparisons are still meaningful.

Table 1 shows a comparison of our method against various prior systems. In order to better isolate the effect of using mel spectrograms as features, we compare to a WaveNet conditioned on linguistic features[8] with similar modifications to the WaveNet architecture as introduced above. We also compare to the original Tacotron that predicts linear spectrograms and uses Griffin-Lim to synthesize audio, as well as concatenative [30] and parametric [31] baseline systems, both of which have been used in production at Google. We find that the proposed system significantly outpeforms all other TTS systems, and results in an MOS comparable to that of the ground truth audio. †

System	MOS
Parametric	3.492 ± 0.096
Tacotron (Griffin-Lim)	4.001 ± 0.087
Concatenative	4.166 ± 0.091
WaveNet (Linguistic)	4.341 ± 0.051
Ground truth	4.582 ± 0.053
Tacotron 2 (this paper)	$\mathbf{4.526 \pm 0.066}$

Table 1. Mean Opinion Score (MOS) evaluations with 95% confidence intervals computed from the t-distribution for various systems.

We also conduct a side-by-side evaluation between audio synthesized by our system and the ground truth. For each pair of utterances, raters are asked to give a score ranging from -3 (synthesized much worse than ground truth) to 3 (synthesized much better than ground truth). The overall mean score of -0.270 ± 0.155 shows that raters have a small but statistically significant preference towards ground truth over our results. See Figure 2 for a detailed breakdown. The comments from raters indicate that occasional mispronunciation by our system is the primary reason for this preference.

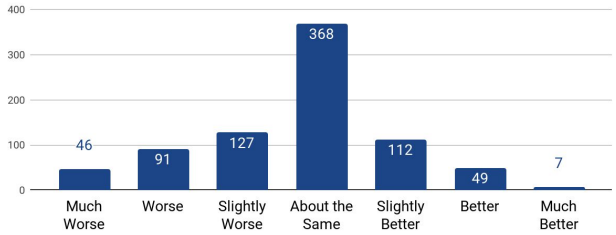

Fig. 2. Synthesized vs. ground truth: 800 ratings on 100 items.

We ran a separate rating experiment on the custom 100-sentence test set from Appendix E of [11], obtaining a MOS of 4.354. In a manual analysis of the error modes of our system, counting errors in each category independently, 0 sentences contained repeated words, 6 contained mispronunciations, 1 contained skipped words, and 23 were subjectively decided to contain unnatural prosody, such as emphasis on the wrong syllables or words, or unnatural pitch. End-point prediction failed in a single case, on the input sentence containing the most characters. These results show that while our system is able to reliably attend to the entire input, there is still room for improvement in prosody modeling.

Finally, we evaluate samples generated from 37 news headlines to test the generalization ability of our system to out-of-domain text. On this task, our model receives a MOS of 4.148 ± 0.124 while WaveNet conditioned on linguistic features receives a MOS of 4.137 ± 0.128. A side-by-side evaluation comparing the output of these systems also shows a virtual tie – a statistically insignificant preference towards our

† Samples available at https://google.github.io/tacotron/publications/tacotron2.

results by 0.142 ± 0.338. Examination of rater comments shows that our neural system tends to generate speech that feels more natural and human-like, but it sometimes runs into pronunciation difficulties, e.g., when handling names. This result points to a challenge for end-to-end approaches – they require training on data that cover intended usage.

3.3. Ablation Studies

3.3.1. Predicted Features versus Ground Truth

While the two components of our model were trained separately, the WaveNet component depends on the predicted features for training. An alternative is to train WaveNet independently on mel spectrograms extracted from ground truth audio. We explore this in Table 2.

	Synthesis	
Training	Predicted	Ground truth
Predicted	4.526 ± 0.066	4.449 ± 0.060
Ground truth	4.362 ± 0.066	4.522 ± 0.055

Table 2. Comparison of evaluated MOS for our system when WaveNet trained on predicted/ground truth mel spectrograms are made to synthesize from predicted/ground truth mel spectrograms.

As expected, the best performance is obtained when the features used for training match those used for inference. However, when trained on ground truth features and made to synthesize from predicted features, the result is worse than the opposite. This is due to the tendency of the predicted spectrograms to be oversmoothed and less detailed than the ground truth – a consequence of the squared error loss optimized by the feature prediction network. When trained on ground truth spectrograms, the network does not learn to generate high quality speech waveforms from oversmoothed features.

3.3.2. Linear Spectrograms

Instead of predicting mel spectrograms, we experiment with training to predict linear-frequency spectrograms instead, making it possible to invert the spectrogram using Griffin-Lim.

System	MOS
Tacotron 2 (Linear + G-L)	3.944 ± 0.091
Tacotron 2 (Linear + WaveNet)	4.510 ± 0.054
Tacotron 2 (Mel + WaveNet)	$\mathbf{4.526 \pm 0.066}$

Table 3. Comparison of evaluated MOS for Griffin-Lim vs. WaveNet as a vocoder, and using 1,025-dimensional linear spectrograms vs. 80-dimensional mel spectrograms as conditioning inputs to WaveNet.

As noted in [10], WaveNet produces much higher quality audio compared to Griffin-Lim. However, there is not much difference between the use of linear-scale or mel-scale spectrograms. As such, the use of mel spectrograms seems to be a strictly better choice since it is a more compact representation. It would be interesting to explore the trade-off between the number of mel frequency bins versus audio quality in future work.

3.3.3. Post-Processing Network

Since it is not possible to use the information of predicted future frames before they have been decoded, we use a convolutional post-processing network to incorporate past and future frames after decoding to improve the feature predictions. However, because WaveNet already contains convolutional layers, one may wonder if the post-net is still necessary when WaveNet is used as the vocoder. To answer this question, we compared our model with and without the post-net, and found that without it, our model only obtains a MOS score of 4.429 ± 0.071, compared to 4.526 ± 0.066 with it, meaning that empirically the post-net is still an important part of the network design.

3.3.4. Simplifying WaveNet

A defining feature of WaveNet is its use of dilated convolution to increase the receptive field exponentially with the number of layers. We evaluate models with varying receptive field sizes and number of layers to test our hypothesis that a shallow network with a small receptive field may solve the problem satisfactorily since mel spectrograms are a much closer representation of the waveform than linguistic features and already capture long-term dependencies across frames.

As shown in Table 4, we find that our model can generate high-quality audio using as few as 12 layers with a receptive field of 10.5 ms, compared to 30 layers and 256 ms in the baseline model. These results confirm the observations in [9] that a large receptive field size is not an essential factor for audio quality. However, we hypothesize that it is the choice to condition on mel spectrograms that allows this reduction in complexity.

On the other hand, if we eliminate dilated convolutions altogether, the receptive field becomes two orders of magnitude smaller than the baseline and the quality degrades significantly even though the stack is as deep as the baseline model. This indicates that the model requires sufficient context at the time scale of waveform samples in order to generate high quality sound.

Total layers	Num cycles	Dilation cycle size	Receptive field (samples / ms)	MOS
30	3	10	6,139 / 255.8	4.526 ± 0.066
24	4	6	505 / 21.0	4.547 ± 0.056
12	2	6	253 / 10.5	4.481 ± 0.059
30	30	1	61 / 2.5	3.930 ± 0.076

Table 4. WaveNet with various layer and receptive field sizes.

4. CONCLUSION

This paper describes Tacotron 2, a fully neural TTS system that combines a sequence-to-sequence recurrent network with attention to predicts mel spectrograms with a modified WaveNet vocoder. The resulting system synthesizes speech with Tacotron-level prosody and WaveNet-level audio quality. This system can be trained directly from data without relying on complex feature engineering, and achieves state-of-the-art sound quality close to that of natural human speech.

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